**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 028 · Common Network Protocols

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is an IPv4 address used for and how is it usually written?

Q2. What is a router's main job?
A. Store user passwords
B. Print network documents
C. Run spreadsheet formulas
D. Link different networks

Q3. “The leftmost bit is always the least significant bit.” Correct this. Explain why.

Q4. Tablets need to join a wired LAN. “A WAP changes domain names into IP addresses.” Correct this. How does a WAP help the tablets?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Protocol — Rules for communication between devices.

HTTP — Hypertext Transfer Protocol, transfers web resources.

HTTPS — Secure HTTP with encrypted communication.

FTP — File Transfer Protocol, transfers files.

SMTP — Protocol used for sending email.

POP — Email-retrieval protocol often downloading mail locally.

IMAP — Email protocol for accessing and synchronising server-based email.

## New learning

Protocols are rules: Communication protocols are agreed rules controlling how data is exchanged across a network.

Web and file protocols: HTTP transfers web pages; HTTPS uses encryption for secure web communication; FTP is used for file transfer.

Email protocols: SMTP sends email; POP downloads email to a client; IMAP accesses and synchronises server-based mail across devices.

TCP and IP: TCP helps ensure reliable, ordered delivery. IP supports addressing and routing so packets reach the intended destination.

## Worked example

A teacher sends an email on a laptop, then reads received messages on a phone. Explain the roles of SMTP and IMAP. [4 marks]

1. Sending and accessing messages are two separate jobs.

2. SMTP is used to send outgoing email [1] from client towards email server [1].

3. IMAP accesses mail stored on a server [1], allowing messages to be synchronised across devices [1].

4. Name each protocol and link it to the specific email action.

# We Do · Practise with guidance

### Guided 1

A student visits a website requiring login details. Explain why HTTPS is preferable to HTTP. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: HTTPS carries web data [1] using encryption [1], making captured traffic unreadable without the key [1] and better protecting sensitive information during transfer [1].

**Guided 2**

A student uses an email app on several devices. Explain the difference between POP and IMAP. [4 marks]

Worked method: POP retrieves/downloads mail to a client [1], often for local storage [1]. IMAP keeps/accesses mail on the server [1] and synchronises changes across multiple devices [1].

**Guided 3**

Describe the different roles of TCP and IP in sending data. [4 marks]

Worked method: TCP helps ensure reliable delivery [1] in the correct order [1]; IP supplies addressing [1] and routing to the destination across networks [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Predict the message.

```
protocol = "HTTPS"
if protocol == "HTTPS" then
    print("Secure web protocol")
endif
```

predict

### Question 1

State one function of SMTP. [1 mark] Source: Adapted from your Paper 1 revision booklet, 1.3.2.

### Question 2

A charity accepts donations on a public website. Explain how HTTPS differs from HTTP and why that matters when supporters enter payment information. [4 marks] Source: Original J277-style question; not an OCR past-paper original.

### Question 3

Describe the uses of FTP, SMTP, IMAP, TCP and IP. [6 marks] Source: Original J277-style question; not an OCR past-paper original.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Exit Ticket · Answer before checking

SMTP is mainly used to send emails.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. IPv4 supports addressing and routing. It is usually written as four decimal numbers separated by dots.

Q2. D — Link different networks

Q3. The MSB is the leftmost bit, weight 128; the LSB is the rightmost bit, weight 1.

Q4. A wireless access point connects wireless devices to a network. It has a different job from a switch or router.

## I Do / We Do pages · Guided practice

Q1. HTTPS carries web data [1] using encryption [1], making captured traffic unreadable without the key [1] and better protecting sensitive information during transfer [1].

Q2. POP retrieves/downloads mail to a client [1], often for local storage [1]. IMAP keeps/accesses mail on the server [1] and synchronises changes across multiple devices [1].

Q3. TCP helps ensure reliable delivery [1] in the correct order [1]; IP supplies addressing [1] and routing to the destination across networks [1].

## You Do pages · Code First

```
protocol = "HTTPS"
if protocol == "HTTPS" then
    print("Secure web protocol")
endif
```

Expected result: Secure web protocol

## You Do pages · Independent questions

Q1. Sends email between clients/servers or mail servers [1].

Q2. HTTP is used to transfer web resources [1]. HTTPS adds encryption to web communication [1], so an interceptor without the appropriate key cannot read captured payment data [1], protecting sensitive information in transit [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q3. FTP transfers files [1], SMTP sends mail [1], IMAP accesses/synchronises received mail [1], TCP checks reliable and in-order delivery [1], IP addresses packets [1] and routes them between networks [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**